$S_0 = 120$

$C_0 = 5 \quad P_0 = 10 \qquad r = 5\%$

$K = \$110 \qquad T = 1$

find an arbitrage $F_{0T} = 100$ (other video $F_{0T} = 110$)

↳ F_{0T}^{NA} from put call parity is \$104.736

- long market forward ✓ – short NA forward

$$-F_{0T}^{P} = -C_0 - \frac{K}{e^{rT}} + P_0$$

short call, short bond, long put

	$t=0$	$t=T=1$	
		$S_t < K$	$S_t > K$
short call	$+C_0 = +5$	0 (OTM)	$-S_T + K$ (ITM) $-S_T + 110$
short bond "borrow"	$+PV(K) = \frac{110}{e^{rT}} = \frac{110}{e^{0.05}}$	$-K = -110$	$-K = -110$
long put	$-P_0 = -10$	$K - S_T = 110 - S_T$	0
	99.63	$-S_T$	$-S_T$
long forward, long bond	0 $-PV(100) = -\frac{100}{e^{0.05}}$	$+S_T - F_{0T} = S_T - 100$ $+100$	
	$+99.63 - \frac{100}{e^{0.05}} \simeq 4.5 > 0$	0	